chapter 6

HCI in the software process

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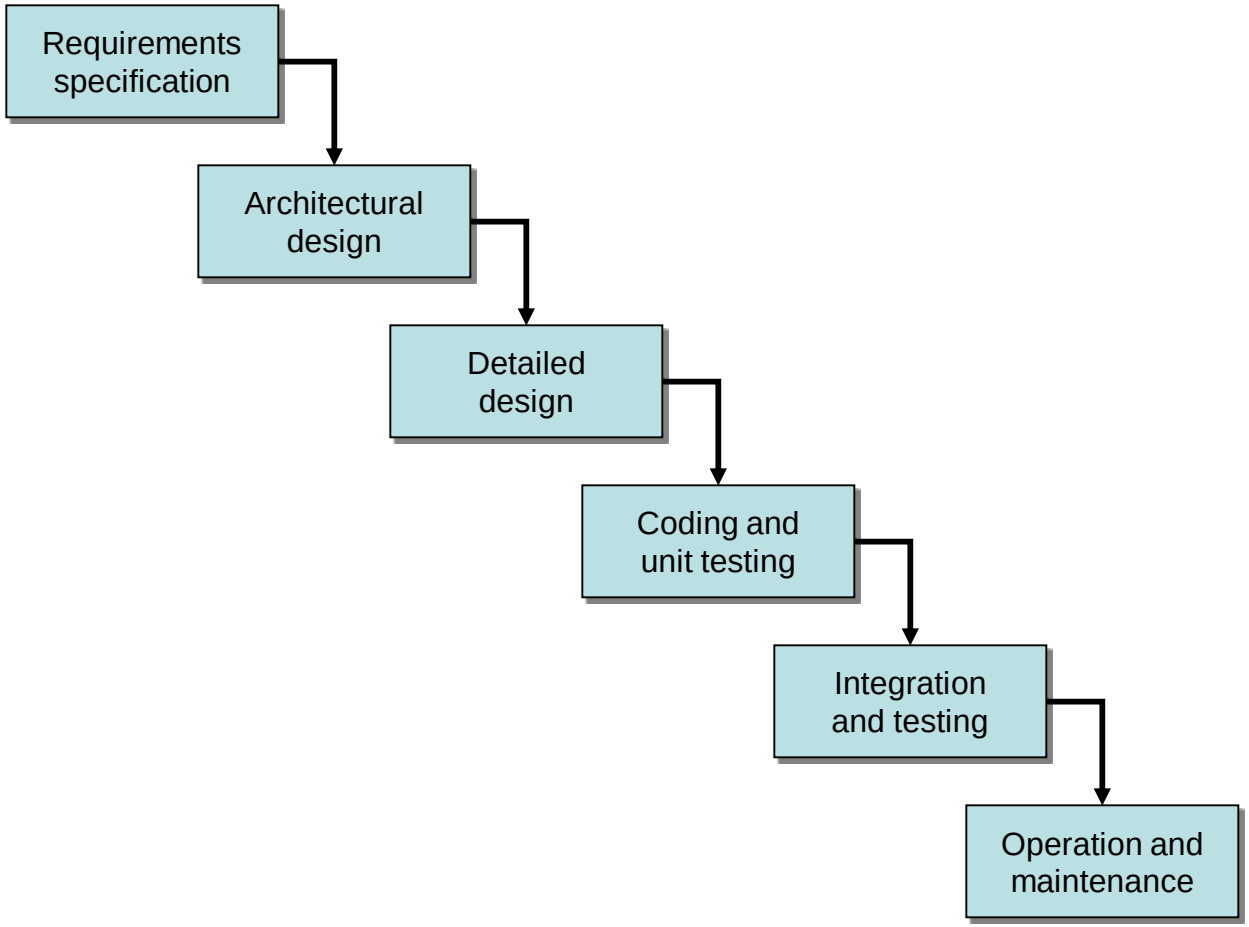
- Software engineering and the design process for interactive systems
- Usability engineering
- Iterative design and prototyping
- Design rationale

the software lifecycle

- Software engineering is the discipline for understanding the software design process, or life cycle

- Designing for usability occurs at all stages of the life cycle, not as a single isolated activity

The waterfall model



Activities in the life cycle

Requirements specification

- designer and customer try capture what the system is expected to provide can be expressed in natural language or more precise languages, such as a task analysis would provide

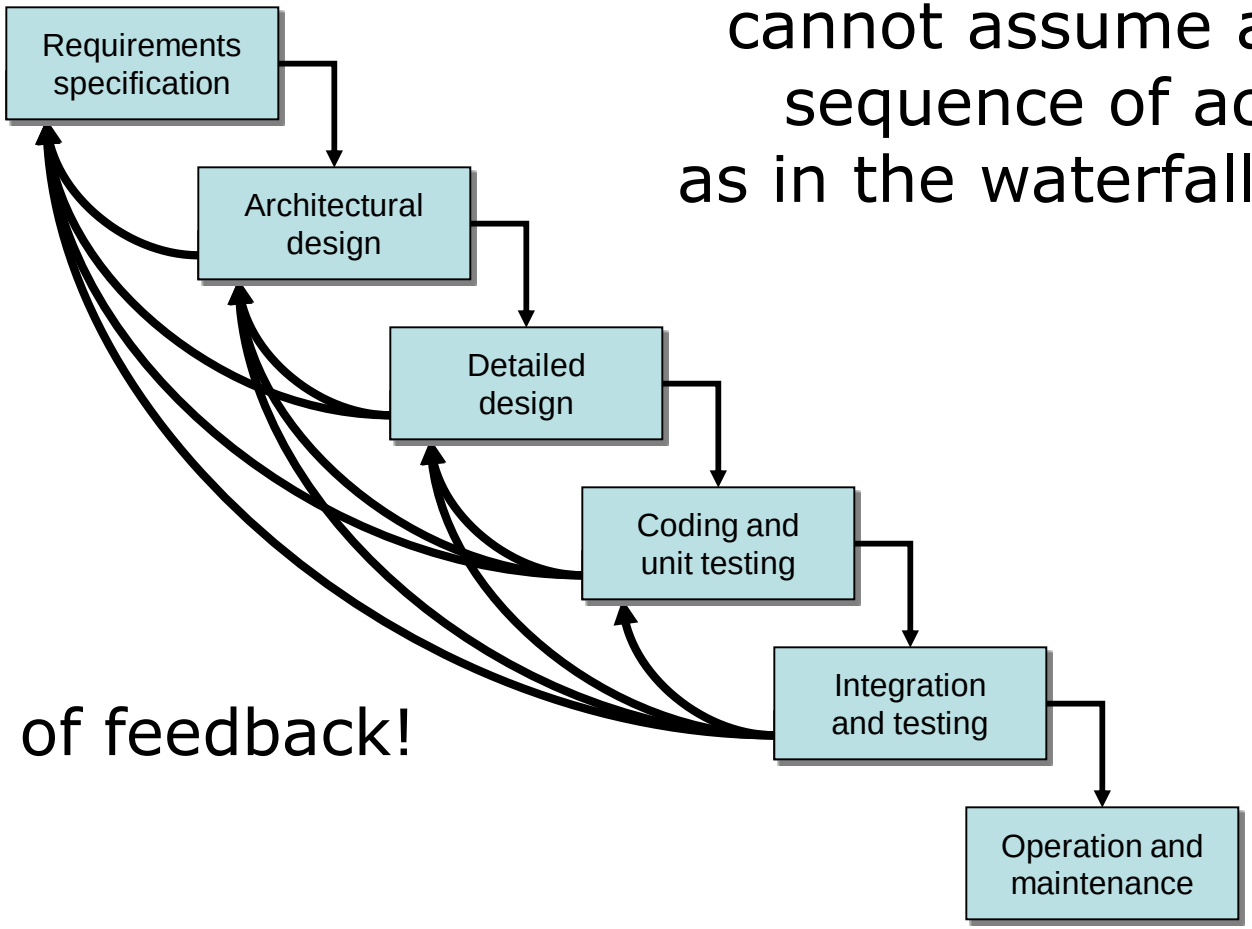
Architectural design

- high-level description of how the system will provide the services required factor system into major components of the system and how they are interrelated needs to satisfy both functional and nonfunctional requirements

Detailed design

- refinement of architectural components and interrelations to identify modules to be implemented separately the refinement is governed by the nonfunctional requirements

The life cycle for interactive systems



cannot assume a linear sequence of activities as in the waterfall model

lots of feedback!

Usability engineering

Usability specification defines clear and measurable requirements for how easy and effective a system should be to use.

Main components:

- Usability attribute: such as efficiency or learnability.
- Measuring concept: What will be measured, such as task completion time or number of errors.
- Measuring method: How the measurement will be collected, such as observing users performing a task.
- Usability levels:
  - Current level: Existing performance.
  - Worst-case level: The least acceptable performance.
  - Planned level: The target performance.
  - Best-case level: The best expected performance.

ISO usability standard 9241

adopts traditional usability categories:

- effectiveness
  - can you achieve what you want to?
- efficiency
  - can you do it without wasting effort?
- satisfaction
  - do you enjoy the process?

Iterative design and prototyping

- Iterative design overcomes inherent problems of incomplete requirements

- Prototypes
  - simulate or animate some features of intended system
  - different types of prototypes
    - throw-away (paper , sketch,.. etc.)
    - Incremental (step by step)
    - Evolutionary (improve, increase features)

- Management issues
  - time
  - non-functional features

Techniques for prototyping

Storyboards

A storyboard is a sequence of drawings that shows how a user interacts with a system. It does not need to be computer-based.

Limited functionality simulations

A limited functionality simulation is a prototype that looks or behaves like the real system, but only some functions are implemented.

Tools like HyperCard are common for these

Wizard of Oz technique

users interact with a prototype as if it were a real system, but a human secretly performs some of the system's functions.

Design rationale

Design rationale explains why a system was designed in a particular way.

For example, a designer may place the registration button on the home page because students use it frequently. A designer places the search button at the top of a website because users need to find information quickly.

Benefits of design rationale

- It helps team members understand design decisions.
- Designers learn from previous projects.
- It records the reasons for choosing one solution over another.
- It helps future developers understand the system and improve it.

Design rationale (cont'd)

Types of DR:

- Process-oriented
  - record the discussion step by step, from the first question to the final decision.
- Structure-oriented
  - Organises design alternatives and their reasons.
  - Focuses on comparing options and explaining the final choice.
- Two examples:
  - Issue-based information system (IBIS)
  - Design space analysis

Issue-based information system (IBIS)

- basis for much of design rationale research
- process-oriented
- main elements:
  - issues
    - hierarchical structure with one 'root' issue
  - positions
    - potential resolutions of an issue
  - arguments
    - modify the relationship between positions and issues

Design space analysis

- structure-oriented
- QOC – hierarchical structure:
  - questions (and sub-questions)
    - represent major issues of a design
  - options
    - provide alternative solutions to the question
  - criteria
    - the means to assess the options in order to make a choice

Summary

The software engineering life cycle
- distinct activities and the consequences for interactive system design

Usability engineering
- making usability measurements explicit as requirements

Iterative design and prototyping
- limited functionality simulations and animations

Design rationale
- recording design knowledge
- process vs. structure